POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Simulation and Modeling (New)

Semester: Spring

Year : 2026
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What are simulation and modelling? Explain traffic management system simulation with their components. 7
   b) Explain the steps in Simulation study with help of flowchart. 8
2. a) Use monte Carlo method to compute $\int_3^5 4x^2 + 6\,dx$ using 10 samples. Aso estimate the error percentage. 7
   b) What is CSSLs? Explain about the structure of CSMP III for automobile suspension system. 8

OR

Write CSMP-III program for Predator-Prey model.

3. a) Explain concept of liver model. Demonstrate it using analog model with the components of analog computer. 7
   b) What is difference between occupancy and utilization? Why do we calculate it? Explain it using an example. 8

OR

Explain different types of time advancement mechanism? Give examples representing the types.

4. a) When line is busy or link is not available, call cannot be connected and the call gets lost. Explain the various states involving each condition. 7
   b) Consider the sequence of random numbers: 0.31, 0.45, 0.63, 0.27, 0.56, 0.81, 0.74, 0.39. Perform Run up and run down test to test for independence at 5% level of significance. $Z_{0.025} = 1.96$. 8
5. a) Using the Chi-Square Test at the 5% significance level ($\alpha = 0.05$), test whether the following random numbers are uniformly distributed. 7

0.15 0.82 0.46 0.71 0.23
0.94 0.38 0.67 0.11 0.57
0.29 0.88 0.75 0.49 0.06
0.91 0.35 0.53 0.80 0.19
0.62 0.41 0.97 0.26 0.73
0.58 0.13 0.84 0.32 0.69

   b) How would you use replication of runs to calculate means, variances, and confidence intervals of output paramater? 8
6. a) Define SIMSCRIPT program. Explain SIMSCRIPT program for inventory management system. 7
   b) What is initial bias? What are various methods to eliminate initial bias? 8
7. Write short notes on: **(Any two)** 2×5
   a) Estimation Methods
   b) Interactive System
   c) Generation of Arrival pattern

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